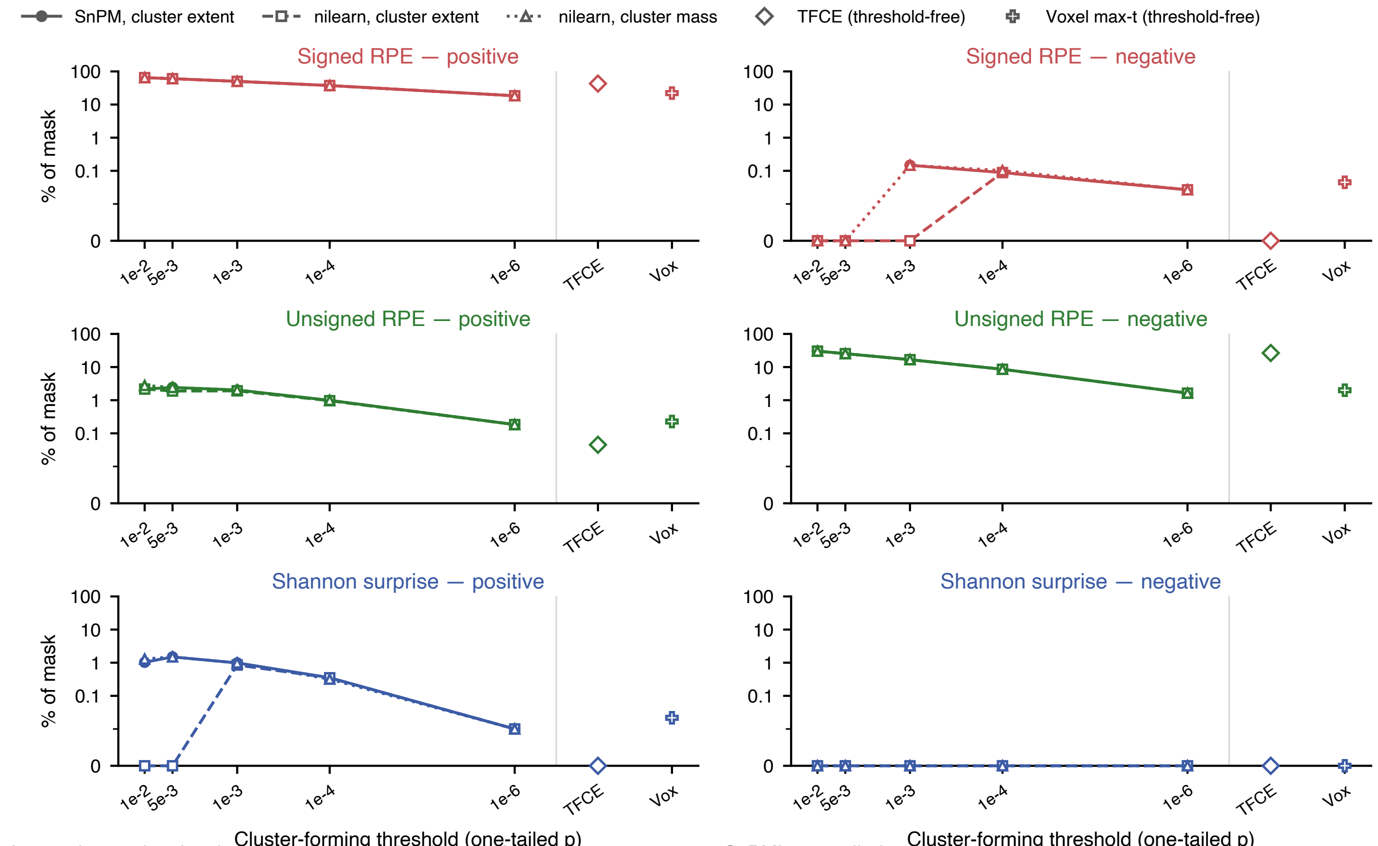
Both engines are permutation tests — what actually differs is the family and the statistic

model7 · n = 58 · 5000 permutations each · same contrast images, same mask (59,838 voxels), nilearn patched to 18-connectivity



Across the 23 signal × sign × threshold panels where both engines return a map, 16 agree to the voxel (Dice > 0.995). Where they disagree it is always a marginal effect, and the cause is the family, not the software:

Signed RPE neg, $p < 0.001$: SnPM 88 vox vs nilearn 0
Shannon surprise pos, $p < 0.01$: SnPM 616 vox vs nilearn 0
Shannon surprise pos, $p < 0.005$: SnPM 894 vox vs nilearn 0

SnPM's one-tailed-per-tail family is the more permissive of the two.

The Shannon-surprise clusters sit right on the boundary: their nilearn two-tailed cluster-EXTENT p is .071 and .054 — not significant — while cluster MASS gives .048 and .021, and SnPM's one-tailed family puts them under .05. Same data, three defensible procedures, opposite verdicts.

The RPE and uRPE maps are indifferent to all of this. The surprise map is not, and neither is the small uRPE-positive set.